

Recognizing Operator Task States from Multimodal Physiological Signals in Industrial Human–Robot Collaboration: a Subject-Disjoint Evaluation

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Abstract. Collaborative and assistive robots that share a workspace with a human operator must react not only to the geometry of the task but to the state of the person performing it: whether the operator is cognitively loaded, stressed, or engaged in a demanding physical manipulation changes what kind of assistance is useful and when it should be offered. Physiological signals, such as brain activity through Electroencephalography (EEG) and autonomic arousal through Electrocardiography (ECG), Electrodermal Activity (EDA), Electromyography (EMG), and respiration, provide a continuous, non-intrusive window onto these states. Yet a core challenge prevents such signals from driving adaptation in practice: most reported evaluations do not transfer to new operators. Windows of the same operator are commonly split at random between training and test, recording metadata leaks into the feature matrix, and the between-operator variability that dominates real deployments is left unquantified. We propose, in this paper, a leakage-free, strictly subject-disjoint evaluation of task-state recognition on the public MultiPhysio-Human–Robot Collaboration (HRC) dataset. Results show that gradient boosting achieves the best across-operator macro-F1 of 0.466 ± 0.149 , statistically tied with eXtreme Gradient Boosting (XGBoost), while within-subject classifiers on the identical features reach 0.91, an operator-induced gap of roughly 0.45 points that dwarfs every other factor in the error budget. Fusing EEG and peripheral features consistently and significantly outperforms the best single modality, and all detectors classify a window in at most $\sim 90 \mu\text{s}$, confirming real-time feasibility.

Keywords: physiological signals · task-type recognition · multimodal fusion · human-robot collaboration · subject-disjoint evaluation · reproducibility

1 Introduction

In shared and assistive HRC, a robot and an operator work on the same physical task and must coordinate in real time. Coordinating well requires the robot to interpret not only the geometric state of the shared workspace but also the state of the person working in it: whether an operator is under high mental load, stressed, or performing a demanding physical manipulation changes what kind of assistance is useful and how it should be timed [13]. Physiological signals offer a continuous and non-disruptive window onto these states, such as brain activity through EEG and autonomic arousal through ECG, EDA, EMG, and respiration, and are increasingly used to drive adaptive behaviour in collaborative robotics [6,3].

The task we address is *task-state recognition*: given a window of physiological signals, decide which kind of activity the operator is engaged in. We deliberately use coarse, task-condition classes, namely a cognitive-load battery, a stressor, an industrial physical task, rest, and a residual class, rather than fine-grained continuous states, because coarse classes are the actionable unit for an assistive policy and because they are exactly what the recording protocol can certify without invasive labelling. Yet even this deliberately simple formulation is hard in practice, for two reasons that shape everything that follows. *Operator dependence*: physiological baselines and response magnitudes differ sharply between individuals, so a model trained on one set of operators can appear excellent on a random held-out subset of *the same* operators’ windows while failing on a genuinely new operator. Any evaluation that does not separate operators risks reporting a ceiling rather than a generalization result. *Class imbalance*: in the data we use, one class, industrial task, accounts for 63% of windows while two minority classes together account for 5%, which makes aggregate accuracy uninformative without class-level controls.

Recognition of physiological operator state has been pursued along three main lines, each of which leaves a distinct gap open. Subject-state methods train on questionnaire-labelled stress and cognitive-load reports and consistently observe that recognition across operators is hard, but they inherit the subjectivity and the coverage of self-reports [5,6,3,1]. Fusion methods argue that central and peripheral modalities carry complementary information, yet the argument is mostly taxonomic and rarely tested under a controlled design on data where channels are heavily redundant [2]. Evaluation-centred reviews have shown that the design of the data split is the single largest driver of reported performance, but they stop at cataloguing the heterogeneity rather than supplying a clean protocol [11,9].

In spite of these advances, some issues still exist. Firstly, evaluations of physiological windows are frequently not subject-disjoint: repeated, auto-correlated

windows from the same operator and the same condition are split randomly between training and test, and metadata such as recording identifiers leak into the feature matrix. The resulting numbers are flattering and irreproducible. Secondly, the between-operator variability that practitioners actually face is rarely quantified: a reader of most papers cannot tell what fraction of the reported accuracy reflects recognition of the *task* and what fraction reflects recognition of the *person*, because no within-subject upper bound is reported alongside the across-operator result. Thirdly, fusion of central and peripheral physiology is generally asserted to help, but the claim is rarely tested with a modality ablation under the identical cleaning and protocol.

This paper addresses these gaps by proposing a leakage-free, strictly subject-disjoint benchmark of task-state recognition on the public MultiPhysio-HRC dataset. We first audit the exported feature table and remove 112 columns that are recording metadata or byte-identical duplicated EEG exports, leaving 128 physiological features, and we label each window deterministically from its protocol condition into one of five task-condition classes. We then evaluate eleven off-the-shelf detectors and one neural fusion architecture under a Leave-One-Subject-Out (LOSO) protocol over all 52 operators, refitting every preprocessing step inside each training fold, and we keep a within-subject run on the identical features only as an explicit upper bound. The headline finding is honest and, we believe, useful: the best detector reaches a per-fold macro-F1 of 0.466 ± 0.149 across new operators, whereas within-subject classifiers on the same features reach 0.91, a generalization gap of roughly 0.45 points that dwarfs the differences between models. A modality ablation under the same protocol shows that fusing EEG with peripheral signals adds a small but significant gain over the best single modality, and that explicit two-tower fusion does not beat plain concatenation. Finally, all detectors classify a window in at most a few tens of microseconds, so real-time assistance is not constrained by inference cost. The contributions are:

1. a **leakage-free feature-cleaning protocol** for a public physiological-HRC dataset, together with a machine-readable manifest recording, for every feature, its modality group and its keep/drop decision with the reason, and a deterministic five-class task-condition labelling rule.
2. a **strictly subject-disjoint LOSO benchmark** of eleven classical and neural detectors over all 52 operators, with preprocessing refit inside each fold, per-fold score files, paired significance tests, and calibration reporting, alongside a within-subject run kept as an explicit upper bound.
3. a **modality-contribution analysis** under this protocol that answers, under a controlled design, whether fusion of central and peripheral physiology actually helps.
4. a quantitative statement of the **generalization gap** between within-subject and across-operator performance, together with real-time latency figures, a deployment and privacy discussion, and a released reproducibility package.

The rest of this paper is structured in the following manner. Section 2 reviews related work. Section 3 describes the dataset, the feature audit, and the

label schema. Section 4 details the models and the LOSO evaluation protocol. Section 5 reports results and Section 6 discusses them and the threats to validity. Section 7 concludes with implications and future directions.

2 Related Work

Physiological operator-state recognition for HRC has seen significant advances through datasets, fusion models, and evaluation studies. This section outlines recent contributions in each of these areas while highlighting the unique drawbacks that motivate the present work.

Bussolan et al. [5] introduced the MultiPhysio-HRC dataset used in this paper and reported subject-disjoint recognition of *self-reported* stress and cognitive-load levels on an industrial assembly task, reaching macro-F1 ranges of roughly 0.34–0.39 for stress and 0.33–0.49 for cognitive load across conditions [6]. These modest numbers, obtained with questionnaire ground truth under a subject-disjoint protocol, are an important reference point: they indicate that across-operator physiological recognition is genuinely hard and that honest protocols produce honest results. However, their targets come from STAI-Y1/NASA-TLX self-reports that are not part of the public feature export, the protocol covers a narrower assembly setting, and the source study neither audits its feature export for leakage nor tests whether fusing modalities helps under a controlled design.

Borghi et al. [3] assessed operator stress in collaborative robotics with a multimodal approach and likewise insist on careful subject-independent evaluation, reporting large variance across individuals. Aygun et al. [1] estimated cognitive workload for assistive robots from physiological signals and again observed pronounced between-subject variability. These works establish that the operator effect is the central difficulty, but they evaluate single-target states on single-architecture pipelines, so the relative contribution of features, fusion, and evaluation design remains entangled.

On the fusion side, Baltrusaitis et al. [2] provide the standard taxonomy of multimodal machine learning, arguing that heterogeneous streams carry complementary information and that naive fusion can hurt when modalities are redundant or noisy. The taxonomy is largely descriptive: controlled *empirical* tests on physiological-HRC data, where redundancy across channels is high and the fusion benefit is not guaranteed, are scarce. Our modality-contribution analysis is a step toward filling that gap.

On the evaluation side, Iarlori et al. [9] survey cognitive-workload estimation from wearables in human-machine interaction and catalogue the evaluation heterogeneity that makes studies hard to compare. Jin et al. [11] provide a recent systematic review of physiological cognitive-state recognition and show that the design of the data split is the single largest driver of reported performance differences. These reviews identify the disease but do not supply the cure: neither proposes a concrete, reproducible subject-disjoint protocol with an explicit within-subject upper bound.

Finally, Selvaggio et al. [13] survey autonomy levels in physical human–robot interaction and identify reliability, transparency, and adaptation as the constraints any state estimator must serve. Task-state recognition is a building block of such autonomy, and our latency and failure-mode analysis speaks to the deployment requirements they enumerate.

In summary, prior work establishes that fusion of physiological modalities is promising in principle [2], that operator-state recognition in HRC is hard and subject-dependent [5,6,3,1], and that evaluation design determines reported performance [11,9]. What is missing is a compact, reproducible, subject-disjoint benchmark of task-condition recognition on a public HRC dataset that simultaneously audits the features, tests whether fusion helps, and quantifies the within-subject versus across-operator gap. The present work directly addresses this gap.

3 Dataset and Task Formulation

3.1 MultiPhysio-HRC

We use the public MultiPhysio-HRC dataset of multimodal physiological recordings collected during controlled and realistic industrial human–robot collaboration [5]. It comprises 52 healthy operators who each performed a battery of 21 protocol conditions while wearing a multi-sensor setup: an EEG headset plus peripheral sensors for ECG, EDA, EMG, and respiration. From the continuous recordings the dataset publishes fixed-duration signal windows together with statistical descriptors. The export we obtained contains 5,640 windows, each originally carrying 257 columns.

3.2 Auditing the feature export

Before any modelling, we audited the exported feature table for the two most common leakage sources in this kind of data.

First, *recording-metadata leakage*. Sixteen columns aggregate the recording identifier, `ID_*`, and the trial counter, `Repetition_*`. These identify the recording session and its order, not the operator’s physiology: they are removed, because they would otherwise let a model short-circuit on trial metadata. Second, *duplicated EEG features*. The export contains each EEG descriptor twice, once under a clean name and once under a `eeg_features_5s_...` prefix, byte-identical. The 96 duplicated columns are removed so that EEG is weighted exactly once.

The audit is summarized in Table 1. After cleaning, the 128 remaining physiological features are statistical descriptors, namely mean, standard deviation, median, and energy, of windowed signals. A machine-readable manifest recording the decision and the reason for every column is released with the reproducibility package.

Table 1. Feature-export audit. The 257 exported columns reduce to 128 physiological features after removing 16 recording-metadata columns and 96 byte-identical duplicated EEG descriptors. The label columns are retained as keys. No model input is taken from them.

Column group	Count	Disposition
Physiological features, EEG	96	kept
Physiological features, ECG	8	kept
Physiological features, EDA	8	kept
Physiological features, EMG	8	kept
Physiological features, respiration	8	kept
EEG duplicated descriptors	96	dropped, exact copies
Recording metadata, <code>ID_*</code> and <code>Repetition_*</code> ,	16	dropped, leakage

3.3 Task-condition labels

MultiPhysio-HRC annotates each recording with the *protocol condition* that was performed and with self-report questionnaires. The questionnaires are not part of the public feature export we obtained, so we do not claim to recognize self-reported affective state. Instead, we define the recognition target as the *task condition*, which is certified by the experimental protocol itself: each window receives exactly one of five labels, assigned deterministically from the recording’s `task_name` by the rule in Table 2, with the rule released verbatim in the reproducibility package. Because the labels are a deterministic function of the protocol, they are exact and reproducible: no crowdsourcing, no subjective rating, no ambiguity about which condition a window came from.

The five classes are chosen so that every protocol condition maps to a class with a coherent physiological hypothesis.

- **cognitive load:** 975 windows, a sustained-attention and memory battery, namely Stroop easy/hard, N-back, Tower of Hanoi, and arithmetic/puzzle. These conditions tax working memory and executive control.
- **high stress:** 224 windows, a virtual-reality height-plank walking task, a validated stressor inducing threat-related autonomic arousal.
- **industrial task:** 3,571 windows, ten physical assembly and order-picking conditions, five performed manually and five with a collaborative robot. These mix physical and cognitive demands in realistic HRC settings.
- **low load:** 799 windows, rest and guided-meditation segments between conditions, providing a low-demand baseline.
- **other:** 71 windows, a single open-ended virtual-reality job-simulation session that belongs to none of the controlled batteries. It is kept as its own class rather than forced into another, and its small size is reported explicitly.

The task is therefore *well-posed but not trivial*: the five classes correspond to different conditions, yet physiological differences across individuals are large enough that a classifier must separate *classes* from *persons*. This is precisely the separation that a subject-disjoint protocol enforces, described in Section 4.5.

Table 2. Deterministic mapping from the 21 protocol conditions to the five task-condition classes, with window counts. Every row of the cleaned data is labelled by this rule. No crowdsourcing or manual labelling is involved.

Class	Protocol conditions, <code>task_name</code>	Windows
cognitive load	<code>stroopeasy_0</code>	138
	<code>stroophard_0</code>	98
	<code>n-back_0</code>	204
	<code>hanoi_0</code>	337
	<code>mat_0</code>	198
high stress	<code>vr-plank_0</code>	224
	<code>manual-task-1...-5</code>	1,587
	<code>cobot-task-1...-5</code>	1,984
	<code>rest-1,rest-2,rest_0</code>	654
	<code>meditation_0</code>	145
low load	<code>vr-job-sim_0</code>	71
industrial task		
	Total	5,640

4 Methodology

4.1 Pipeline overview

The processing and evaluation pipeline is shown in Figure 1. A single preprocessing path produces the fused tabular representation used by every detector. Modality feature groups are kept separate alongside it so that the modality-contribution analysis of Section 5 reuses the identical path. All detectors share one evaluation protocol, so any difference in outcome is attributable to the detector itself rather than to the data preparation.

4.2 Problem formulation

Let $\{x_i, y_i, s_i\}_{i=1}^N$ denote the cleaned windows, where $x_i \in \mathbb{R}^{128}$ is the fused feature vector, $y_i \in \mathcal{Y} = \{1, \dots, 5\}$ is the task-condition label, with cognitive load = 1, high stress = 2, industrial task = 3, low load = 4, and other = 5, and $s_i \in \{1, \dots, 52\}$ is the operator who produced the window. Two statistical facts about this data drive the design of the objective. Windows of the same operator are not independent, and repeated windows of the same condition are strongly autocorrelated. The class counts of Table 2 are heavily imbalanced, with one majority class covering 63% of the windows.

For a model f_θ producing class posteriors $p_\theta(y = c \mid x_i)$, we train with the class-weighted cross-entropy objective

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{c=1}^5 w_c \mathbf{1}[y_i = c] \log p_\theta(y = c \mid x_i), \quad (1)$$

Task-condition recognition from multimodal physiological signals

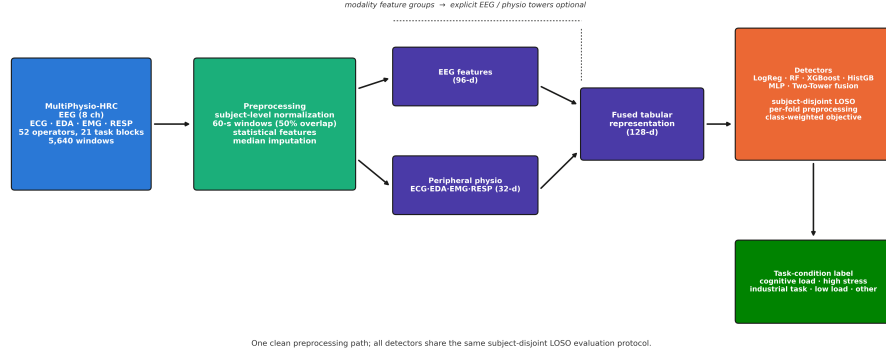


Fig. 1. Processing and evaluation pipeline. One clean preprocessing path feeds a fused 128-dimensional tabular representation. The same path exposes modality feature groups for the fusion ablation. All detectors are evaluated under the identical subject-disjoint LOSO protocol with preprocessing refit inside each fold.

where $\mathcal{B}$ is a training batch, the full set for batch models, and $w_c \propto N/(5 N_c)$ is the inverse-frequency weight of class c with $N_c = \sum_i \mathbf{1}[y_i = c]$. The weights counteract the imbalance of Table 2 so that the loss is not dominated by the majority *industrial task* class. Without them, a trivial constant classifier already attains 63% accuracy. The decision rule at test time is the maximum-posterior rule

$$\hat{y}_i = \arg \max_{c \in \mathcal{Y}} p_\theta(y = c \mid x_i), \quad (2)$$

which, together with the class weighting in Eq. (1), targets class-balanced behaviour rather than raw accuracy.

4.3 Preprocessing and feature representation

Preprocessing is deliberately minimal and identical across models so that performance differences are attributable to the detectors and to the protocol, not to bespoke feature engineering:

1. columns audited as in Section 3, the 128 kept features forming x_i .
2. missing values are imputed with the training-fold median, and the audit found 328 missing entries out of $5,640 \times 128$.
3. features are standardized to zero mean and unit variance using training-fold statistics only, and
4. a fused representation is used by all tabular detectors, while the modality-feature groups, EEG and peripheral, are exposed to the ablation and to the two-tower architecture.

Crucially, imputation and standardization are *refit inside every training fold*, as described in Section 4.5. Test operators never contribute to the statistics that transform their own windows, which removes a subtle and common form of leakage.

4.4 Detectors

To justify our choice of detectors: the aim is to make the result a property of the representation and the protocol rather than of one model family, so we compare eleven off-the-shelf, lightweight detectors spanning the main families used on tabular physiological features: linear models, instance-based methods, tree ensembles, and one neural network. The eleven detectors are logistic regression with class weighting, k -nearest neighbours, decision tree, random forest, extra-trees, gradient boosting, histogram gradient boosting, XGBoost [7], linear Support Vector Machine (SVM), Gaussian naive Bayes, and an Multilayer Perceptron (MLP), all implemented with scikit-learn [12]. Tree ensembles and boosting are natural baselines for tabular physiological features: XGBoost hyperparameters are fixed a priori, namely 400 trees, depth 6, learning rate 0.05, and column and row subsampling 0.9, rather than tuned on the test operators, which would itself leak information about them.

In addition we evaluate one neural fusion architecture, *two-tower*, which makes the modality split explicit rather than concatenating features at the input. A peripheral tower and an EEG tower each map their feature group to a 64-dimensional embedding, built from two linear-ReLU-dropout layers, the two embeddings are concatenated, and a two-layer head produces the five logits. Two-tower is trained with Adam, using learning rate 10^{-3} and weight decay 10^{-4} , class-weighted cross-entropy Eq. (1), and early stopping on a held-out validation slice inside each training fold. It is the only detector with learnable modality structure: comparing it with fused-tabular XGBoost answers whether an explicit fusion architecture adds anything over plain concatenation.

4.5 Evaluation protocol

The evaluation answers one question: *how well does a detector trained on known operators recognize the task state of a new operator?* We therefore use LOSO cross-validation over the 52 operators. Each operator in turn forms the test fold. The detector and the preprocessing are fitted on the remaining 51 operators' windows only. This yields 52 per-operator evaluations, whose scores are aggregated.

Within one fold, the test windows come from a single operator and are heavily autocorrelated, so the per-fold score is an estimate for that operator, not a collection of independent samples. We therefore treat the 52 per-fold scores as the primary unit of analysis and report their mean and standard deviation, a 95% confidence interval of the mean computed with Student t over folds, and the pooled out-of-fold (OOF) metrics obtained by concatenating every operator's out-of-fold predictions.

Metrics over a fixed label space. Because a fold may contain no window of a minority class, per-fold scores must be computed over the *fixed* five-class space, with a missing class contributing a recall of zero. Averaging only over the classes present in a fold would make scores incomparable across folds. Let the confusion matrix in fold f be $C^f \in \mathbb{N}^{5 \times 5}$. Class recall is $r_c^f = C_{cc}^f / \max(\sum_{c'} C_{cc'}^f, 1)$, and the balanced accuracy is

$$\text{BA}_f = \frac{1}{5} \sum_{c=1}^5 r_c^f, \quad (3)$$

which is the average recall, following Brodersen et al., who define balanced accuracy as mean recall [4]. The macro-F1 is

$$\text{MacroF}_f = \frac{1}{5} \sum_{c=1}^5 \text{F1}(c), \quad \text{F1}(c) = \frac{2p_c r_c}{p_c + r_c}, \quad (4)$$

where p_c, r_c are the class precision and recall in the fixed space and a class absent from both prediction and truth has $\text{F1}(c) = 0$ [14]. Macro-F1 is our primary ranking metric because it penalizes both precision and recall per class and weighs all five classes equally despite the 63%/4%/1% imbalance [10]. We additionally report pooled balanced accuracy, pooled accuracy, and, for probabilistic models, the expected calibration error, ECE, [8] of the pooled OOF probabilities, which quantifies how well a detector’s confidence tracks its correctness: this is needed if downstream assistance will gate on confidence.

Comparisons and upper bound. Every detector is compared with the best per-fold macro-F1 model by a paired Wilcoxon signed-rank test over the 52 folds, with the mean difference and a paired effect size reported. We also run a *within-subject* evaluation on the identical cleaned features: a stratified random 80/20 split in which the same operators appear in both training and test. This is explicitly *not* a claim of generalization: it measures an optimistic ceiling in which the model has already met the operator, and its contrast with the LOSO result quantifies the generalization gap.

All random seeds are fixed, and per-fold, per-class, and pooled result files are written for every model and released. Inference latency is measured as the per-window CPU prediction time of each fitted detector.

5 Results

5.1 Comparison under LOSO

Table 3 reports all eleven detectors plus the two-tower network under the subject-disjoint LOSO protocol, ranked by per-fold macro-F1.

The ranking tells a coherent story. Gradient boosting is the best detector, reaching a per-fold macro-F1 of 0.466 ± 0.149 , with a 95% confidence interval of $[0.42, 0.51]$, over the 52 held-out operators, and XGBoost is statistically tied with

Table 3. Subject-disjoint leave-one-subject-out results on the clean 128-feature set, with 52 operators. Macro-F1 and balanced accuracy are averaged over the 52 held-out subjects and shown as mean $\pm$ std. ECE is computed on pooled out-of-fold probabilities. Latency is the per-window CPU inference time of a model fitted on a training mini-sample. Best value per column in bold.

Model	Macro-F1 mean $\pm$ std	Balanced acc. mean $\pm$ std	Accuracy mean $\pm$ std	ECE	Inference μ s/window
GB	0.466 $\pm$ 0.149	0.470 $\pm$ 0.152	0.690 $\pm$ 0.197	0.028	5
XGBoost	0.462 $\pm$ 0.148	0.478 $\pm$ 0.153	0.654 $\pm$ 0.213	0.127	19
HistGB	0.451 $\pm$ 0.146	0.477 $\pm$ 0.143	0.627 $\pm$ 0.227	0.202	13
RF	0.430 $\pm$ 0.132	0.429 $\pm$ 0.139	0.668 $\pm$ 0.208	0.153	69
SVM-RBF	0.422 $\pm$ 0.149	0.467 $\pm$ 0.158	0.588 $\pm$ 0.217	0.178	51
ExtraTrees	0.421 $\pm$ 0.137	0.421 $\pm$ 0.141	0.658 $\pm$ 0.221	0.167	91
MLP	0.374 $\pm$ 0.160	0.408 $\pm$ 0.161	0.548 $\pm$ 0.232	0.342	2
LogReg	0.371 $\pm$ 0.155	0.415 $\pm$ 0.157	0.552 $\pm$ 0.242	0.215	2
kNN	0.362 $\pm$ 0.118	0.356 $\pm$ 0.119	0.598 $\pm$ 0.207	0.128	6
GaussNB	0.344 $\pm$ 0.133	0.394 $\pm$ 0.140	0.491 $\pm$ 0.204	0.441	20
Tree	0.338 $\pm$ 0.131	0.350 $\pm$ 0.132	0.521 $\pm$ 0.215	0.448	2

it at 0.462 ± 0.148 with paired $p = 0.88$. The rest of the field trails by 0.02–0.13. The top of the table is dominated by tree ensembles, while linear models, k NN, and naive Bayes trail clearly, evidence that the class boundaries in the statistical feature space are not linearly separable, so additive models have no way to isolate the minority classes. Pooled balanced accuracy is consistently well below pooled accuracy, for example gradient boosting reaches accuracy 0.69 versus balanced accuracy 0.47, confirming that the majority class inflates accuracy and that the class-level metrics of Eqs. (3)–(4) are the ones to trust. Calibration is uneven rather than uniformly poor: the best detector is well calibrated, with Expected Calibration Error (ECE) of 0.03, while the other strong models range from 0.13 to 0.45, so confidence-thresholded assistance would need model-specific recalibration before deployment.

The pooled OOF confusion matrix of the best model in Figure 2 shows where the errors concentrate, and Table 4 gives the corresponding class-level metrics. The four substantial classes are recovered with F1-score (F1) between 0.54 and 0.85, ordered as *industrial task* at 0.85, *high stress* at 0.59, *cognitive load* at 0.55, and *low load* at 0.54. The ordering is mechanistically sensible: industrial-task windows come from homogeneous, physically demanding conditions that leave a strong peripheral signature, whereas cognitive-load and low-load classes share an attentional substrate and are therefore the most confusable. The *other* class, 71 windows drawn from a single protocol condition, is never recovered: its windows scatter mostly into *industrial task* and *high stress*. Its zero recall is an honest consequence of the class being one recording session rather than a generalizable physiological state, and it should be read as such rather than as a failure of the features.

Table 4. Pooled out-of-fold per-class metrics of the best LOSO model, GB, over all 5,640 test windows. Each subject contributes only out-of-fold predictions. Window counts are the true-label column totals of the pooled confusion matrix.

True label	Precision	Recall	F1	Windows, n
Cognitive load	0.536	0.556	0.546	975
High stress	0.576	0.612	0.593	224
Industrial task	0.868	0.839	0.854	3571
Low load	0.502	0.586	0.541	799
Other, vr-job-sim	0.000	0.000	0.000	71
Total				5640

Table 5. Generalization gap on the identical clean features. Within-subject is a stratified random 80/20 split, in which the same operators appear in both training and test, an optimistic ceiling, while LOSO holds whole operators out. The gap quantifies how much performance is lost when no data from the target operator is available at training time.

Model	Within-subject	LOSO	Gap
XGBoost	0.910	0.462	+0.448
HistGB	0.911	0.451	+0.459
RF	0.877	0.430	+0.447
MLP	0.846	0.374	+0.472

Figure 3 reveals the second main finding: per-operator macro-F1 varies widely, from 0.15, barely above the 0.2 chance floor for a five-class problem, to 0.78 across the 52 operators. A detector that works well for one operator can be near-useless for another, and this variance, not the mean, is the practical obstacle to assistive deployment. It is also the mechanism behind the generalization gap quantified next, and it motivates the personalization discussion in Section 6.

5.2 The generalization gap

Table 5 and Figure 4 contrast the within-subject ceiling with the LOSO result on the identical features. The within-subject classifiers reach 0.91 macro-F1, with XGBoost at 0.910 and HistGB at 0.911, against 0.45–0.46 across operators for the same models, a gap of roughly 0.45 points.

This gap is the headline finding of the paper and the reason row-level evaluations of physiological windows are misleading: most of the apparent accuracy of a row-level split reflects *recognizing the person*, not the task state. The mechanism is the per-operator physiological signature: baselines and response magnitudes that differ sharply between individuals. An assistance system deployed on a new operator without any calibration data should therefore expect performance near the lower, LOSO numbers, not the within-subject ones.

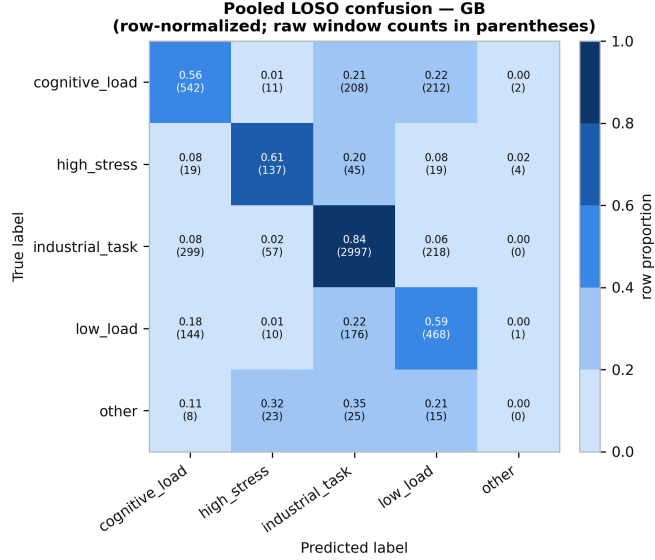


Fig. 2. Pooled OOF confusion matrix of the best LOSO model, row-normalized, with raw window counts. Values are computed by summing the 52 per-operator fold matrices before normalizing.

5.3 Does fusion help?

Table 6 and Figure 5 answer the modality question under the same protocol: for each leading model we run EEG-only, peripheral-only, and fused inputs with 96, 32, and 128 features respectively. Peripheral physiology alone already beats EEG alone by a wide margin, for example XGBoost reaches 0.39 versus 0.33, yet adding EEG on top of the periphery still helps: fused improves over the best single modality for all four models, by +0.072 for XGBoost with $p < 10^{-4}$, +0.050 for HistGB with $p = 0.002$, +0.047 for RF with $p = 0.015$, and +0.037 for MLP with $p = 0.105$, which is not significant. The two-tower network, which fuses through separate modality towers rather than concatenation, reaches 0.406 ± 0.158 , above fused MLP and every single-modality input but significantly below fused XGBoost, by -0.056 with $p = 0.001$.

This result matters for the fusion literature. On this data, the gains from combining central and peripheral physiology are real and statistically significant for the leading models, but they are an order of magnitude smaller than the operator effect that dominates the error budget. In other words, fusion helps, but it is not the bottleneck: an explicit two-tower architecture adds nothing over plain concatenation fed to a strong tree ensemble, a useful negative result for practitioners choosing between simple and structured fusion.

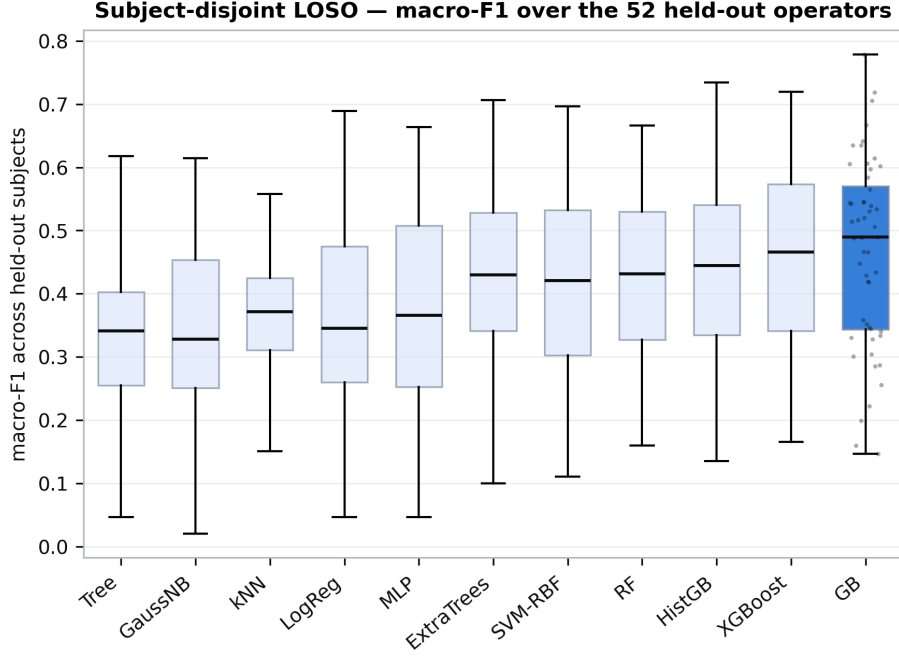


Fig. 3. Distribution of per-operator macro-F1 for every detector under LOSO. Each box summarizes the 52 held-out operators. Large spread means the detector’s usefulness depends strongly on which operator it meets.

5.4 Statistical significance

Figure 6 reports the per-fold paired Wilcoxon comparison of every detector against the best, gradient boosting. Under the across-operator protocol, the top three tree ensembles are statistically indistinguishable: XGBoost at $p = 0.88$, HistGB at $p = 0.38$, while random forest and extra trees separate only at $p = 0.018$ and $p = 0.003$. By contrast, the gap between the tree ensembles and the linear/instance-based/neural group is decisive: every member separates from the best at $p \leq 3 \times 10^{-5}$, and most below 10^{-6} . Together with the effect sizes, this tells the reader which reported differences are worth acting on.

5.5 Real-time feasibility

The sensor windows last on the order of tens of seconds, while all evaluated detectors classify a single window in roughly $2\text{--}90\mu\text{s}$ of CPU time, about $5\mu\text{s}$ for the best detector and $19\mu\text{s}$ for XGBoost, as listed in Table 3. Inference is therefore orders of magnitude faster than real time on commodity CPUs, and real-time adaptive assistance is not constrained by inference cost but by the reliability question of Section 6.

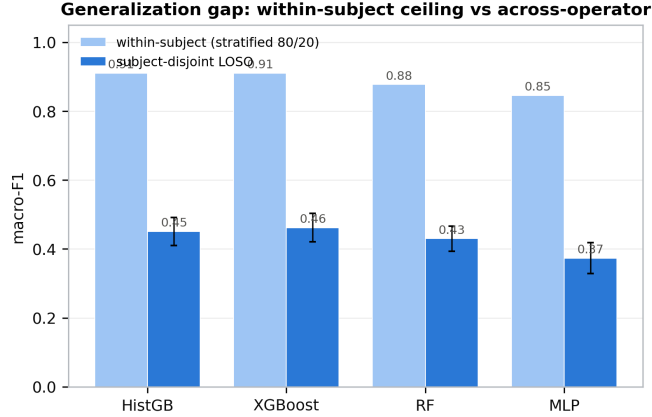


Fig. 4. Within-subject ceiling versus across-operator LOSO macro-F1 on identical features. The gap quantifies how much performance is lost when the target operator was absent from training.

Table 6. Benefit of feature fusion under the identical 52-subject LOSO protocol. Fused 128-dimensional input is compared with the better single modality, EEG-only or physio-only, per model. p -values are paired Wilcoxon signed-rank tests over the 52 folds. Two-Tower is the neural model with explicit per-modality towers, compared with fused XGBoost.

Model	Best single	Single	Fused	Δ	Wilcoxon p
XGBoost	physio	0.390	0.462	+0.072	0.000
HistGB	physio	0.401	0.451	+0.050	0.002
RF	physio	0.383	0.430	+0.047	0.015
MLP	physio	0.337	0.374	+0.037	0.105
Two-Tower <i>vs</i> fused XGB	fused XGB	0.462	0.406	-0.056	0.001

6 Discussion

Taken together, our results reveal three important insights about recognizing operator task states from physiological signals in HRC.

First, the honest across-operator numbers are low, and the gap to the within-subject ceiling is the real result. The best detector reaches 0.466 macro-F1 under LOSO, while the same features with a row-level split report 0.91, a gap we reproduce with a stratified within-subject split and then explain as leakage of operator identity. We consider the LOSO figure the only defensible headline, and we recommend that physiological-recognition papers report the within-subject ceiling and the across-operator result *together*, as in Table 5. Reporting either alone is misleading in opposite directions.

Second, three factors make the problem hard, and none is fixable by a better classifier alone. Firstly, *the operator effect dominates*: physiological baselines

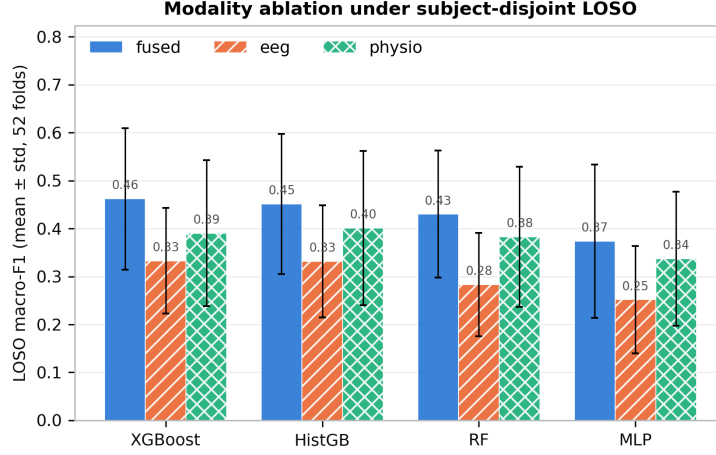


Fig. 5. Modality ablation under LOSO: fused, EEG-only, and peripheral-only inputs with 128, 96, and 32 features respectively for the leading detectors. Error bars are the standard deviation over the 52 folds.

and gains differ so much between people that a model cannot separate “this operator is stressed” from “this operator responds strongly” without calibration data [1,3]. Secondly, *the classes are coarse but confusable*: cognitive-load and industrial-task conditions plausibly share attentional and motor components, so confusion between them is physiologically reasonable. Thirdly, *the imbalance is extreme at the tail*: two minority classes have 224 and 71 windows spread over very few conditions, making them statistically near-impossible to learn across operators.

Third, fusion and speed are not the bottlenecks. Fusing EEG with peripheral signals adds a real but small gain, namely +0.072 macro-F1 for the best model, explicit two-tower fusion does not beat concatenation, and inference takes micro-seconds per window. An assistive system is therefore not limited by what it can compute or how it fuses, but by whether it has met the operator before. This reframing points the research effort where it belongs: at personalization and at calibration of the confidence that would gate assistance.

6.1 Comparison with the source protocol

Bussolan et al. report subject-disjoint recognition of *self-reported* stress and cognitive-load levels with macro-F1 around 0.34–0.49 [5,6]. Our numbers are not directly comparable for two reasons: their targets come from STAI-Y1 / NASA-TLX self-reports on a subset of conditions, whereas ours are deterministic task-condition labels on all conditions, and their protocol concerns a narrower assembly setting. Obtaining the dataset’s questionnaire tables, which are not in the public feature export we used, would let us map our task-condition predictions onto the self-report levels and compare head-to-head. We flag this as

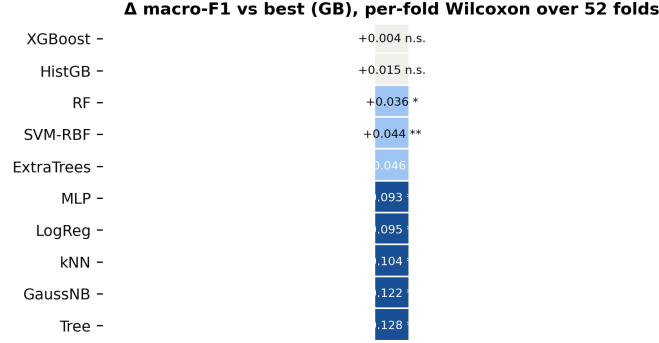


Fig. 6. Mean macro-F1 difference of each detector relative to the best, with the paired Wilcoxon significance over the 52 folds.

the single most valuable extension rather than claim comparability we cannot substantiate.

6.2 Threats to validity

- **Construct validity of labels.** Our labels are task conditions, not measured mental states. A cognitive-load condition may not produce cognitive load in every operator at every window: the label is a *protocol* label. We state this explicitly and never equate a prediction with a diagnosis.
- **Class-size and condition confound.** The *other* class is a single condition, 71 windows, so a detector can effectively memorize that condition. Its recall is reported only for transparency.
- **Within-condition autocorrelation.** Even under LOSO, a test fold contains consecutive windows of the same condition from one operator. Per-fold scores are per-operator estimates, not independent samples. Our paired tests respect this by pairing per-operator scores. The pooling step of Section 4.5 is descriptive, not an independence claim.
- **Feature provenance.** We cleaned the exported feature table we obtained. We did not recompute features from raw signals. The manifest makes the cleaning reproducible, but the upstream feature computation is the dataset authors’.
- **Sensor placement and recording hardware** are fixed by the dataset and were not varied, so results may not transfer to different hardware or real factory noise, where motion artefacts would be larger than in a laboratory HRC setting.

6.3 Deployment considerations

Physiological sensing in the workplace raises privacy and acceptability questions that are not technicalities. EEG headsets and physiological wearables capture

information far beyond task state, and continuous monitoring of stress in industrial settings can be used for worker surveillance rather than assistance. Any deployment should, first, process signals on-device where feasible and retain only task-state decisions, second, use predictions to *adapt robot behaviour*, never for individual performance evaluation, and third, make the operator’s right to opt out and to know what is inferred explicit. A failure/fallback policy is equally important: because confidence is currently unevenly calibrated, as shown in Section 5, assistance must degrade gracefully, falling back to a fixed, safe robot policy whenever the detector’s confidence is low, rather than acting on a low-confidence state estimate.

7 Conclusion and Future Work

Using physiological signals to drive adaptive assistance in industrial HRC requires recognizing what the operator is doing across people, not just within the people a model was trained on. The central difficulty is that physiological baselines and response magnitudes differ sharply between individuals, so an evaluation that does not separate operators reports a ceiling rather than a generalization result, and the between-operator variability that dominates real deployments goes unquantified.

To solve this problem, this paper proposes a leakage-free, strictly subject-disjoint benchmark of task-state recognition on the public MultiPhysio-HRC dataset. The study rests on three components: first, a cleaning protocol that removes recording metadata and duplicated EEG exports, leaving 128 physiological features labelled deterministically into five task-condition classes, second, an evaluation of eleven detectors and one neural fusion network under a LOSO protocol over all 52 operators, with every preprocessing step refit inside each fold, and third, a modality-contribution analysis that tests, under the identical protocol, whether fusing central and peripheral physiology helps.

Our evaluation shows the honest and reproducible picture. The best detector, gradient boosting, reaches a per-fold macro-F1 of 0.466 ± 0.149 across new operators, and within-subject classifiers on the identical features reach 0.91, a generalization gap of roughly 0.45 points that dwarfs the differences between models. Fusion adds a real but small gain, namely +0.072 for the best model, explicit fusion architecture adds nothing over concatenation, and inference takes microseconds per window. These are the properties an assistance system actually faces when it meets an operator for the first time, and they explain why optimistic row-level evaluations of physiological windows do not transfer to new operators.

Future work will pursue three directions. First, *Personalization*: given the operator effect, few-shot calibration, a handful of labelled windows from a new operator, either as adaptation data or as a per-operator prior, is the most direct route to usable accuracy, and our per-operator score files make the potential gain measurable. Second, *Self-report targets*: obtaining the dataset’s questionnaire tables would let us bridge the task-condition schema to the STAI/NASA-

TLX stress and cognitive-load levels used by the source authors, enabling the direct comparison we currently flag as out of scope. Third, *Temporal context*: the present work classifies windows independently. Modelling the condition sequence and per-operator dynamics with recurrent or state-space models is a natural next step, and the latency headroom we measure leaves ample room for such models at inference time.

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